

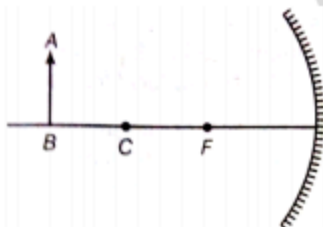
NUMERICAL PROBLEMS IN RAY OPTICS

MANDATORY PROBLEMS

1. Trace the path of a ray of light passing through a glass prism (ABC) as shown in the figure. If the refractive index of glass is $\sqrt{3}$, find out the value of the angle of emergence from the prism.



2. An object AB is kept in front of a concave mirror as shown in the figure.



- (i) Complete the ray diagram showing the image formation of the object.
(ii) How will the position and intensity of the image be affected if the lower half of the mirror's reflecting surface is painted black?
3. The radii of curvature of the faces of a double convex lens are 10 cm and 15 cm. If focal length of the lens is 12 cm, find the refractive index of the material of the lens.
4. A biconvex lens has a focal length $\frac{2}{3}$ times the radius of curvature of either surface. Calculate the refractive index of lens material.
5. Find the radius of curvature of the convex surface of a Plano-convex lens, whose focal length is 0.3 m and the refractive index of the material of the lens is 1.5.
6. A converging lens has a focal length of 20 cm in air. It is made of a material of refractive index 1.6. It is immersed in a liquid of refractive index 1.3. Calculate its new focal-length.
7. An object of 3 cm height is placed at a distance of 60 cm from a convex mirror of focal length 30 cm. Find the nature, position and size of the image formed.
8. The velocity of a certain monochromatic light, in a given transparent medium, is 2.25×10^8 m/s. What is the (i) critical angle of incidence (ii) polarizing angle for this medium?
9. A compound microscope uses an objective lens of focal length 4 cm and eyepiece lens of focal length 10 cm. An object is placed at 6 cm from the objective lens.

Calculate the magnifying power of the compound microscope. Also calculate the length of the microscope.

10. The objective of an astronomical telescope has a diameter of 150 mm and a focal length of 4.00 m. The eyepiece has a focal length of 25.00 mm. Calculate the magnifying and resolving power of telescope. ($\lambda=6000 \text{ \AA}$ for yellow colour).

11. A ray of light is incident at an angle of 60° on one face of a rectangular glass slab of thickness 0.1 m. Calculate the lateral shift produced. Given: Refractive index of glass is 1.5.

12. Find the critical angle for glass-water interface, if the refractive index of glass is $3/2$ and that of water is $4/3$.

13. For a fish in water, outside world would appear to be contained within a circle of area 12.56 m^2 at the surface. If the RI of water is $4/3$, find the depth at which fish is present.

14. An equilateral prism of RI of 1.45 is surrounded by water of RI of 1.33. A narrow beam of light is incident normally on one face of the prism. Find the angle of emergence.

15. A small air bubble within a glass sphere of radius 20 mm and refractive index 1.54 appears to be 10 mm below the surface, when looked along the diameter containing the bubble. Find the real depth of the bubble along with line of sight.

16. A lens has a focal length of 0.2 m when kept in air. Calculate its new focal length when the lens is immersed in water of refractive index 1.33. Given: Refractive index of glass is 1.5.

17. A 2 cm high object is placed on the principal axis of a concave mirror at a distance of 12 cm from the pole. If the image is inverted, real and 5 cm high, find the location of image and focal length of the mirror.

18. A 4.5 cm needle is placed 12 cm away from a convex mirror, of focal length 15 cm. Give the location of the image & magnification. Describe what happens when the needle is moved away from the mirror.

19. A narrow beam of light is incident normally on one face of equilateral glass prism of R.I 1.45. When the prism is immersed in a suitable liquid, the ray makes a grazing emergence along the other face. Find the R.I of this liquid and draw a diagram showing the path of image.

20. A white light passed through a glass prism of angle 5° . If the refractive indices for red & blue colours are 1.641 & 1.659, calculate the angle of dispersion between these colours.

21. A small telescope has an objective lens of focal length 144 cm and eyepiece of focal length 6 cm. What is the magnifying power of the telescope? What is the separation between the objective and eyepiece?

22. An object is placed 15 cm in front of a convex lens of focal length 10 cm. Find the nature and position of the image formed. Where a concave mirror of radius of

curvature 20 cm should be placed so that the final image is formed at the position of the object itself?

23. Light of wavelength $5000 \text{ \AA}$ propagating in air gets partly reflected from the surface of water. How will the wavelengths and frequencies of the reflected and refracted light be affected?

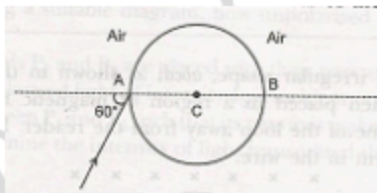
24. (i) A giant refracting telescope has an objective lens of focal length 15 m. If an eyepiece of focal length 1.0 cm is used, what is the angular magnification of the telescope?

(ii) If this telescope is used to view the moon, what is the diameter of the image of the moon formed by the objective lens? The diameter of the moon is $3.48 \times 10^6 \text{ m}$ and the radius of lunar orbit is $3.8 \times 10^8 \text{ m}$.

25. A concave lens of refractive index 1.5 is immersed in a medium of refractive index 1.65. What is the nature of the lens?

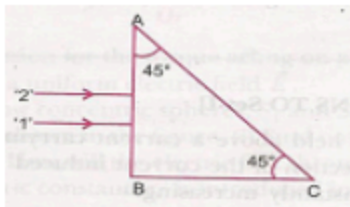
26. A compound microscope has an objective of focal length 1.25 cm and eyepiece of focal length 5 cm. A small object is kept at 2.5 cm from the objective. If the final image formed is at infinity, find the distance between the objective and the eyepiece.

27. A ray of light falls on a transparent sphere with centre C as shown in the figure. The ray emerges from the sphere parallel to the line AB. Find the angle of refraction at A if the refractive index of the material of the sphere is $\sqrt{3}$.



28. A convex lens of focal length 20 cm is placed coaxially with a convex mirror of radius of curvature 20 cm. The two are kept 15 cm apart. A point object is placed 40 cm in front of the convex lens. Find the position of the image formed by this combination. Draw the ray diagram showing the image formation.

29. Two monochromatic rays of light are incident normally on the face AB of an isosceles right-angled prism ABC. The refractive indices of the glass prism for the two rays '1' and '2' are respectively 1.38 and 1.52. Trace the path of these rays after entering through the prism.



30. Where will you keep an object in front of a :

(i) convex lens in order to get a virtual and magnified image?

(ii) concave mirror to get a real and diminished image?

31. The deviation produced for violet, yellow and red lights for crown glass are 3.75° , 3.25° and 2.86° respectively. Calculate the dispersive power of the crown glass.

32. A thin converging lens of focal length 5 cm is used as a simple microscope. Calculate its magnifying power when image formed lies at:

(i) infinity (ii) least distance of distinct vision ($D=25$ cm).

33. A thin converging lens of focal length 12 cm is kept in contact with a thin diverging lens of focal length 18 cm. Calculate the effective/equivalent focal length of the combination.

Sundaram Sir Classes

PRACTICE PROBLEMS

1. Calculate the distance of an object of height, h from a concave mirror of focal length 10 cm, so as to obtain a real image of magnification 2.
2. A screen is placed 100 cm from an object. The image of the object on the screen is formed by a convex lens at two different locations, separated by 20 cm. Calculate the focal length of the lens used.
3. A convex lens made up of glass of refractive index 1.5 is dipped, in turn, in
(i) a medium of refractive index 1.65 (ii) a medium of refractive index 1.33
(a) Will it behave as a converging lens or a diverging lens in the two cases?
(d) How will its focal length change in the two media?
4. The image obtained with a convex lens is erect and its length is four times the length of the object. If the focal length of the lens is 20 cm, calculate the object and image distances.
5. A convex lens is used to obtain a magnified image of an object on a screen 10 cm from the lens. If the magnification is 19, find the focal-length of the lens.
6. (i) How is the focal length of a spherical mirror affected when the wavelength of the light used is increased?
(ii) A convex lens has 20 cm focal length in air. What is its focal length in water? (Refractive index of air-water = 1.33, refractive index of air-glass = 1.5).
7. An object of 2 cm height is placed at a distance of 30 cm from a convex mirror of focal length 15 cm. Find the nature, position and size of the image formed.
8. A tank is filled with water to a height of 12.4 cm. The apparent depth of a needle lying at the bottom of the tank is measured by a microscope to be 9.3 cm. What is the refractive index of water? If water is replaced by a liquid of refractive index 1.6 up to the same height, by what distance would the microscope have to be moved to focus on the needle again?
9. A convex lens of focal length 25 cm is placed coaxially in contact with a concave lens of focal length 20 cm. Determine the power of the combination. Will the system be converging or diverging in nature?
10. Two convex lenses of focal length 20 cm and 1 cm constitute a telescope. The telescope is focused on a point which is 1 m away from the objective. Calculate the magnification produced and the length of the tube, if the final image is formed at a distance, 25 cm from the eyepiece.
11. A small telescope has objective lens of focal length of 140 cm and eyepiece of focal length 5 cm. What is the magnifying power of telescope when
(i) normal adjustment
(ii) final image is formed at least distance of distinct vision.
12. A person with a normal near point using compound microscope with objective of focal length 8 mm and eyepiece of focal length 2.5 mm can bring an object

placed at 9 mm from the objective to sharp focus. What is the separation between the 2 lenses? Calculate the magnifying power of the microscope.

13. The focal length of the objective and the eyepiece of a microscope are 2 cm & 5 cm respectively and distance between them is 20 cm. Find the distance of the object from the objective when the final image seen by the eye is 25 cm from the eyepiece. Also find the magnifying power.

14. An angular magnification of 30X is desired using an objective of focal length of 1.25 cm and eyepiece of focal length 5 cm. How will you set up the compound microscope?

15. The far point of a myopic eye is 10 cm from the eye. What must be the focal length of a lens for reading at normal distance (25 cm)?

16. A convex lens of focal length $f = 20$ cm is combined with a diverging lens of power 6.5 D. What are the power and focal length of the combination?

17. The refractive indices of water, diamond and glass are 1.33, 2.0 and 1.5 respectively. In which of these will be speed of light be the maximum? Justify your answer with a suitable reason.

18. The angle of an equilateral glass prism is 59.5° . The angle of minimum deviation is 38.5° . Without using the value of refractive index of the glass, calculate the angle of incidence for a ray of light suffering minimum deviation.

19. The refractive indices of a prism for violet, yellow and red colours are 1.632, 1.620 and 1.613 respectively. Calculate the dispersive power of the material of the prism.

20. What are the two possible values of the magnifying power of a simple microscope made of convex lens of focal length $f = 5.0$ cm?

21. A ray of light incident in air for a certain air-glass surface of separation has a polarizing angle of 56° . What will be the angle of refraction in glass?

22. The dispersive power of an achromatic doublet is $2/3$ and the combined focal length of the combination is 30 cm. Find the focal length of each lens stating its nature.

23. An astronomical telescope set for normal adjustment has a magnifying power 10. If the focal length of the objective is 1.20 m, what is the focal length of the eyepiece?

24. A biconvex spherical lens has focal length 20 cm. If it is split into two halves to form two Plano-convex lenses, what is the focal length of each Plano convex lens?

25. Two thin lenses, one convex and the other concave of focal length 15 cm and 20 cm respectively, are put in contact. Find the focal length of the doublet lens.

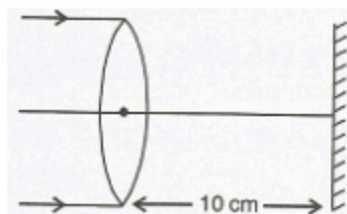
26. A fish is located at a distance of 10 cm from the wall of fish pond. The thickness of the glass wall is 2 cm. Find the apparent position of the fish.

27. A Plano-convex lens made of glass ($\mu = 1.5$) has a focal length 10.0 cm.

(i) Calculate the radius of curvature of its curved surface.

(ii) If an identical Plano-convex lens is placed in contact with the first (plane surfaces facing each other), what will be the power of the combination?

28. A convex lens and a plane mirror are kept at a distance of 10 cm as shown in fig. Parallel rays incident on the convex lens after reflection from the mirror form an image at the optical centre. Complete the ray diagram and find out the focal length of the lens.



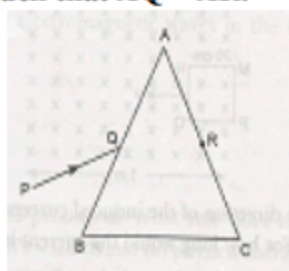
29. A biconvex lens made of a transparent material of refractive index 1.5 is immersed in water of refractive index 1.33. Will the lens behave as a converging or a diverging lens? Give reason.

30. The total magnification produced by a compound microscope is 20. The magnification produced by the eyepiece is 5. The microscope is focussed on a certain object. The distance between the objective and eyepiece is observed to be 14 cm. If least distance of distinct vision is 20 cm, calculate the focal length of the objective and the eyepiece.

31. An equiconvex lens of focal length 'f' is cut into two identical plano convex lenses. How will the power of each part be related to the focal length of the original lens?

32. A double convex lens of +5D is made of glass of refractive index 1.55 with both faces of equal radii of curvature. Find the value of its radius of curvature.

33. A ray PQ incident on the face AB of a prism ABC, as shown in the figure, emerges from the face AC, such that AQ = AR.



Draw the ray diagram showing the passage of the ray through the prism. If the angle of the prism is 60° and refractive index of the material of the prism is $\sqrt{3}$, determine the values of angle of incidence and angle of deviation.